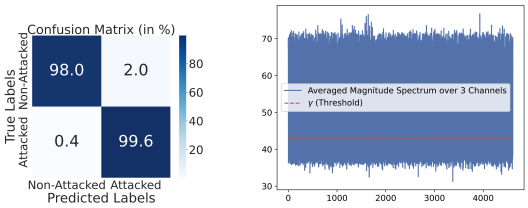
Stealthy Adversarial Image Detection and Denoising on Autonomous Vehicles using a Diffusion-based Model

Bijay Gaudel* and Mohammad Bahrami*



(a) (b)

Fig. 1: Accuracy and Decision-making report. (a) Confusion matrix, (b) Illustration of the averaged magnitude spectrum over 3 channels and the threshold used for classification problem in (2).

(a) (b)

Fig. 2: Prediction of YOLO v5. (a) adversarial attacked image (448×448), (b) Corresponding denoised image (128×128).

The reliability and trustworthiness of machine learning algorithms are of significant importance to many safety-critical applications such as autonomous vehicles. The vulnerability of neural networks to various adversarial perturbations is well documented [1], [2]. There exists a vast literature on methods for adversarial robustness that is to defend against the generated stealthy adversarial perturbations (i.e., inputs imperceptible to human eyes) causing misclassification by the targeted neural network [1]. Some examples are Perceptual Adversarial Training (PAT), gradient masking, and adversarial purification [3], [1]. The prevailing state-of-the-art (SoTA) methods only consider classification tasks in static settings where adversarial attacks run independently of the targeted system's dynamics. Aligned with the recent studies of adversarial attacks in dynamic settings [2], we consider the detection of adversarial image attacks targeting the vision-based navigation of autonomous vehicles. Our framework leverages the generative capability of diffusion probabilistic models for adversarial purification (i.e. noise removal from images) as well as a spectral analysis of the denoised images using Fourier transform for binary decision-making between attacked and non-attacked status.

The goal of diffusion probabilistic models is that for any given image data frame $x_0 \in \mathbb{R}^{H\times W\times 3}$ of a complex distribution $q(x_0)$, we are interested in learning a model distribution $p_\theta(x_0)$ that approximates $q(x_0)$, enabling a generative capability of the trained network that will be used to denoise (purify) the possibly attacked images x_0. To this end, we restrict all of the models to Gaussian distributions and a sequence of latent variables, $\{x_1, x_2, \ldots, x_T\}$, with increasing levels of noise $\epsilon \sim \mathcal{N}(0, \boldsymbol{I})$, are obtained from a forward (diffusion) process that follows the form $x_t = \sqrt{\alpha_t}x_0 + \sqrt{1-\alpha_t}\epsilon$, parameterized by a decreasing sequence $\alpha_{1:T} \in [0, 1)$ such that α_T is sufficiently close to 0. Then, a reverse (generative) process $p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \hat{\mu}_\theta(x_t), \hat{\Sigma}_\theta(x_t))$, with a Unet architecture parameterized by θ and the mean $\hat{\mu}_\theta(x_t)$ and variance $\hat{\Sigma}_\theta(x_t)$ are learned. The training objective function is:

$$\mathcal{L}^t(\theta) = \mathbb{E}_{x_0,\epsilon}\left[\|x_0 - \hat{x}_\theta(\alpha_t x_0 + \sqrt{1-\alpha_t}\epsilon)\|_2^2\right], \quad (1)$$

The decision-making about whether the image is attacked will be performed by analyzing the residual image defined as $\tilde{x}_0 = x_0 - \mathrm{D}(x_0)$, where $\mathrm{D}(\cdot)$ is the trained diffusion model. We use the following classification problem during runtime

$$\mathrm{H}_0 : \text{attack-free}, \qquad \text{if } \mathrm{m}(\tilde{x}_0) \le \gamma, \quad (2a)$$

$$\mathrm{H}_1 : \text{attacked}, \qquad \text{if } \mathrm{m}(\tilde{x}_0) > \gamma, \quad (2b)$$

in which γ is a threshold and $\mathrm{m}(\tilde{x}_0) = \sum_{H=1}^{128}\sum_{W=1}^{128}\sum_{\mathrm{ch}=1}^{3} f(\tilde{x}_0 \cdot \frac{1}{255})$, where $f(\cdot)$ is the Fourier transformation.

In the experiment, Unet served as the diffusion model backbone. T was set to 1000 diffusion steps, employing a sigmoid noise scheduler and Adam optimizer. The learning rate was 8×10^{-5}, with a batch size of 128 and image size of 128×128. The training dataset contains 5120 clean images, while the testing set contains 2280 and 2328 attack-free and attacked images, respectively. The threshold $\gamma = 43$ is used in (2) with the results shown in Figure 1, indicating 98.8% accuracy. Figure 2 shows an example of YOLO model prediction in an oblation study of the proposed denoising model.

References

[1] N. Papernot, P. McDaniel, A. Sinha, and M. P. Wellman, "Sok: Security and privacy in machine learning," in *2018 IEEE European Symposium on Security and Privacy (EuroS&P)*. IEEE, 2018, pp. 399–414.

[2] H.-J. Yoon, H. Jafarnejadsani, and P. Voulgaris, "Learning when to use adaptive adversarial image perturbations against autonomous vehicles," *IEEE Robotics and Automation Letters*, 2023.

[3] W. Nie, B. Guo, Y. Huang, C. Xiao, A. Vahdat, and A. Anandkumar, "Diffusion models for adversarial purification," *arXiv preprint arXiv:2205.07460*, 2022.

*Equal contribution.

The Authors are with the Department of Mechanical Engineering, Stevens Institute of Technology, Hoboken, NJ 07030, USA, {bgaudel,mbahrami,hjafarne}@stevens.edu.